

Supplementary Material: Instrumental Eigenmode Spectroscopy Using Optical Clocks

1 Per-window response series (representative modes; no exclusions)

Only a small number of representative modes are shown to avoid redundant clones. For each selected mode at $\epsilon = 0.02$, the following are shown for both readout groups: (i) SNR vs window start time with null-shuffle bands, (ii) f_{peak} vs window start time, and (iii) demodulated phase on passing windows only (by definition).

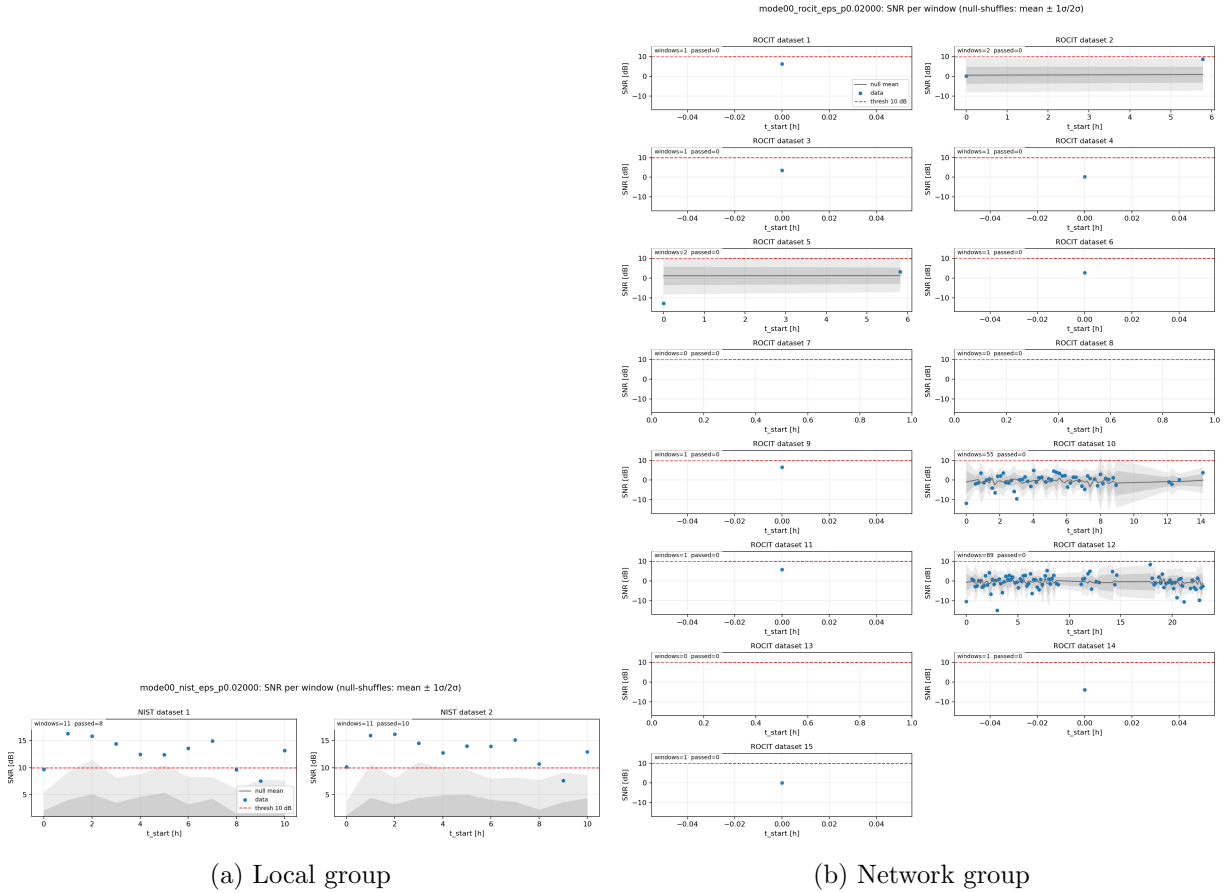


Figure 1: Mode $n = 0$ at $\epsilon = 0.02$: SNR vs window start time. Null-shuffle bands ($\mu \pm \sigma$ and $\mu \pm 2\sigma$) are shown. No windows are excluded.

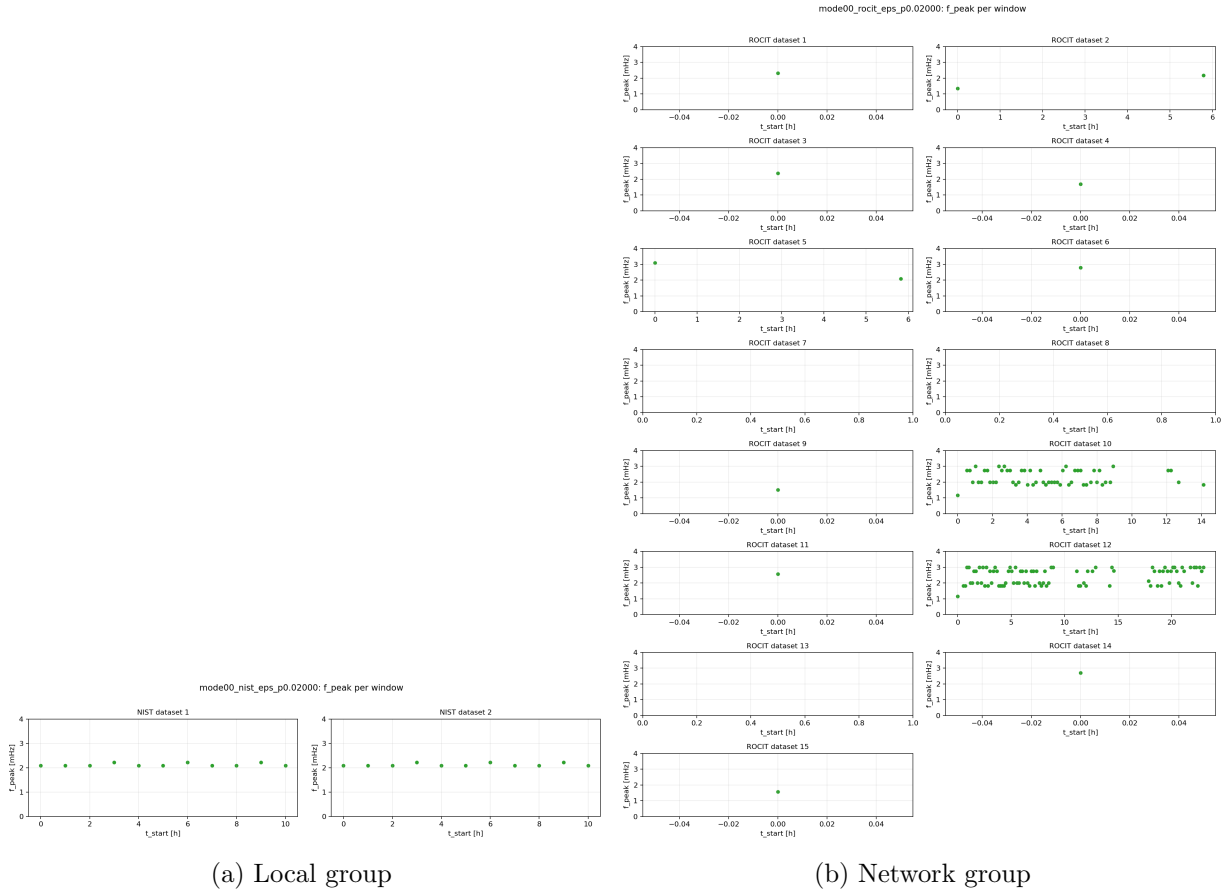


Figure 2: Mode $n = 0$ at $\epsilon = 0.02$: f_{peak} vs window start time.

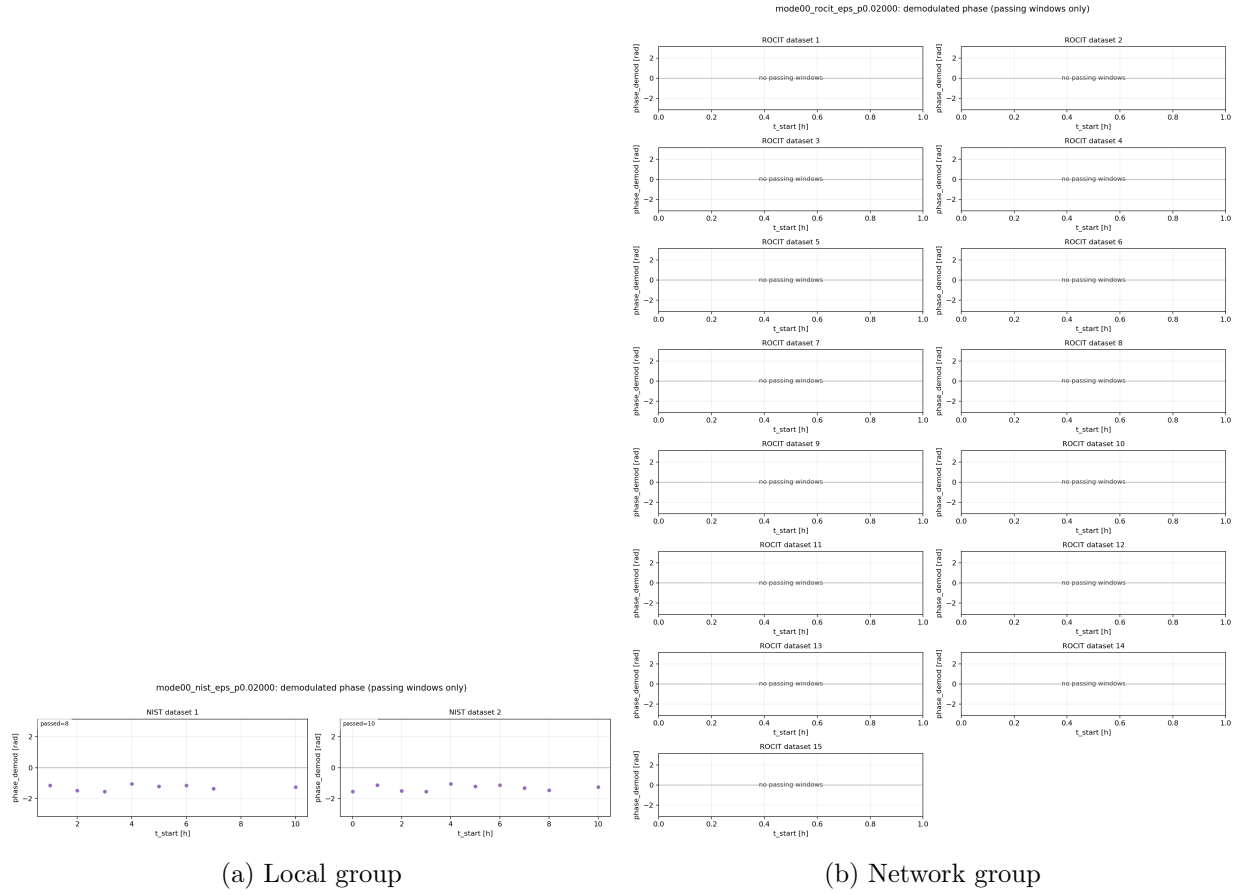


Figure 3: Mode $n = 0$ at $\epsilon = 0.02$: demodulated phase on passing windows only (by definition).

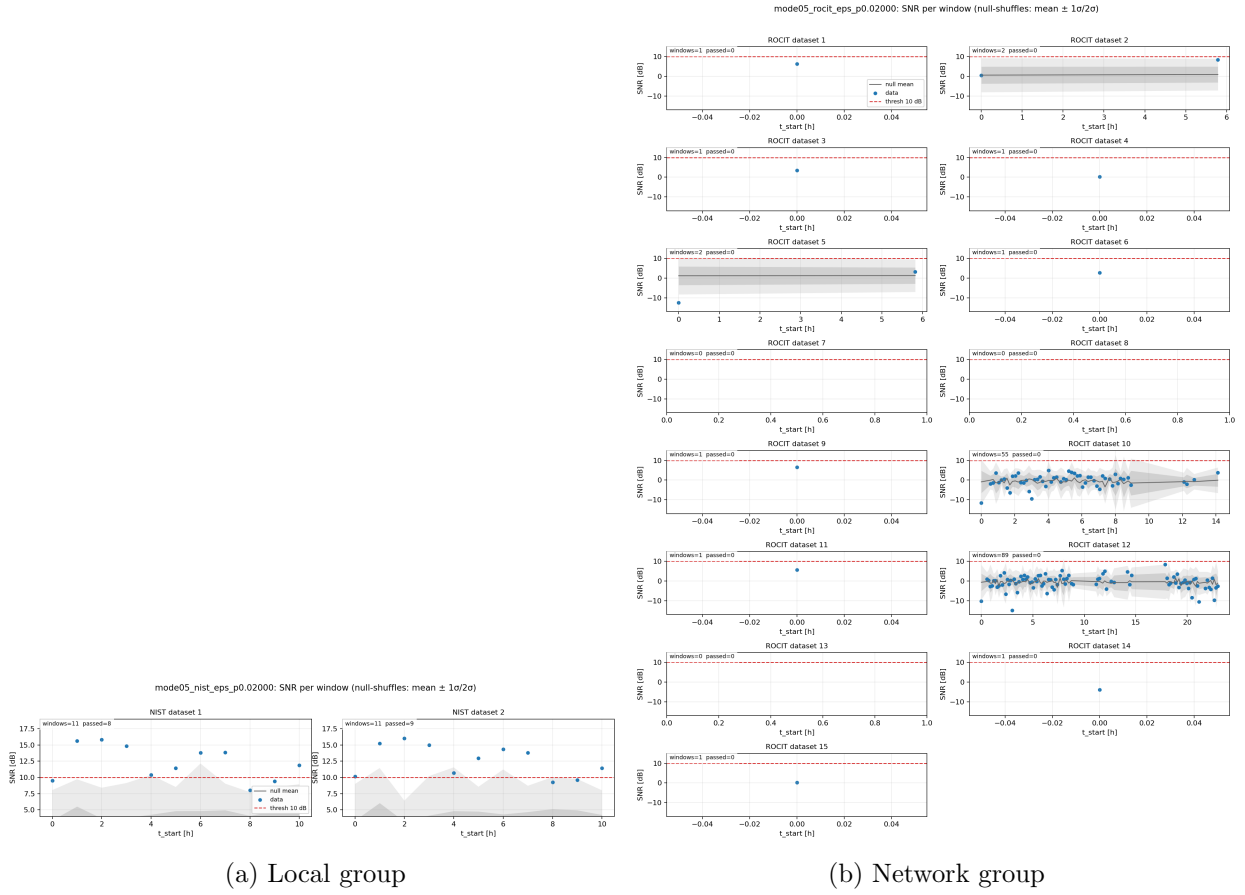


Figure 4: Mode $n = 5$ at $\epsilon = 0.02$: SNR vs window start time. Null-shuffle bands ($\mu \pm \sigma$ and $\mu \pm 2\sigma$) are shown. No windows are excluded.

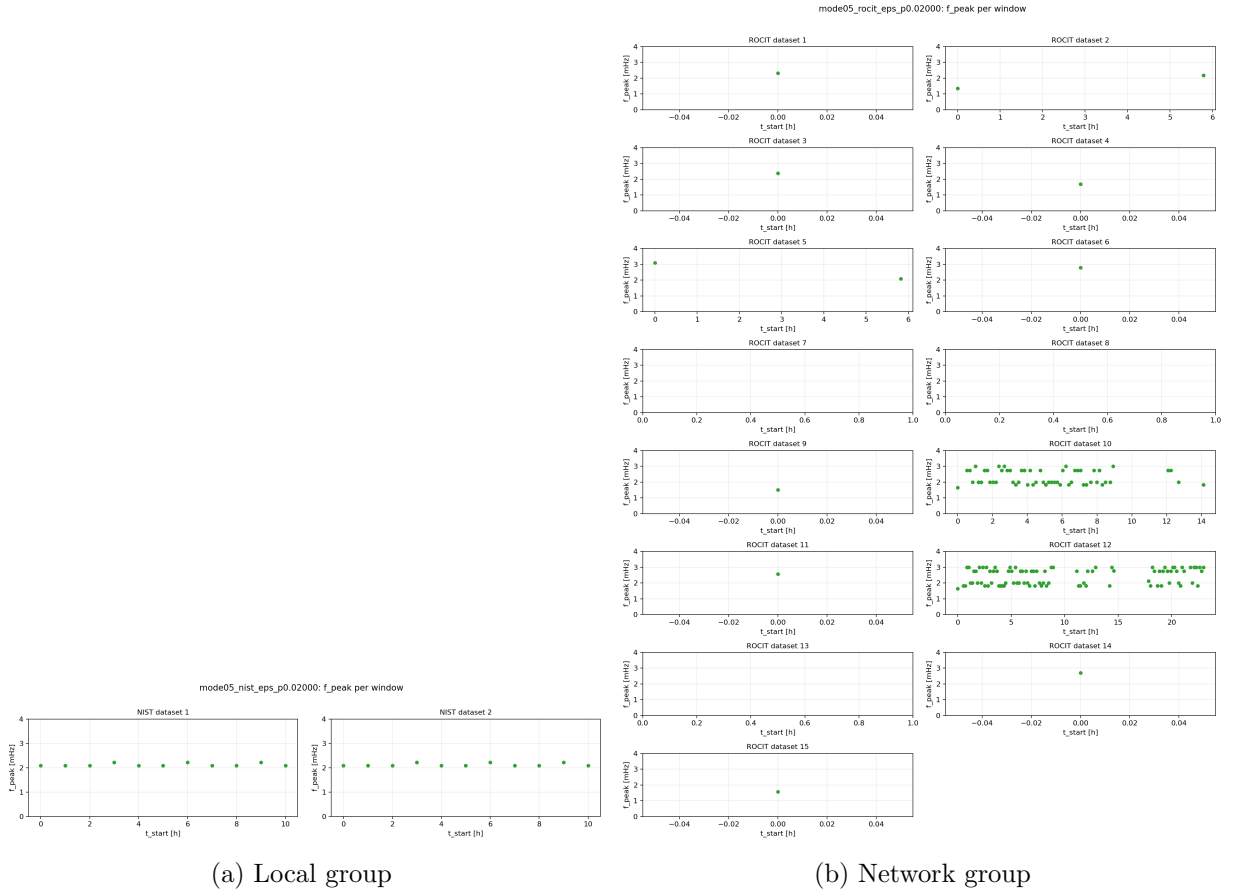


Figure 5: Mode $n = 5$ at $\epsilon = 0.02$: f_{peak} vs window start time.

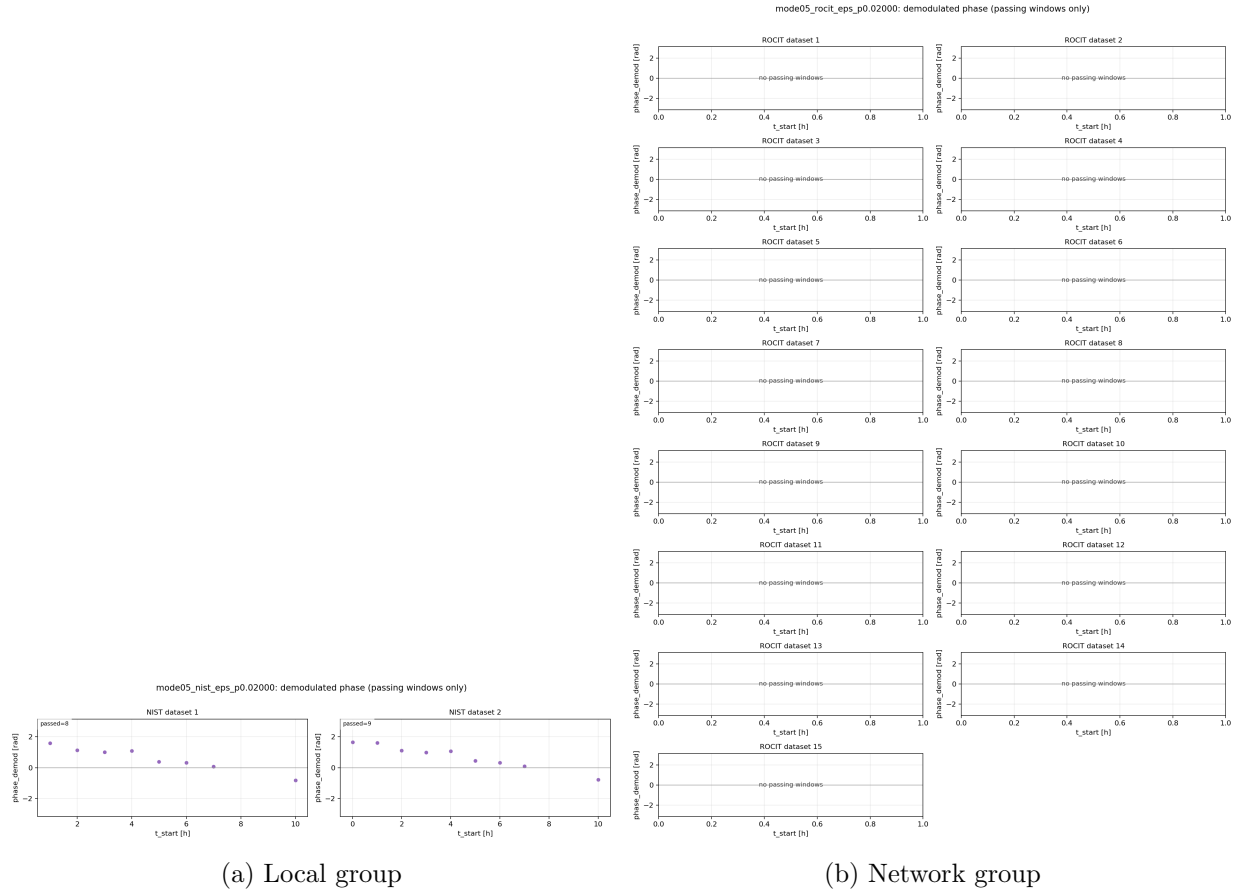


Figure 6: Mode $n = 5$ at $\epsilon = 0.02$: demodulated phase on passing windows only (by definition).